

A Dissociation Test for Causal Attribution Under Partial Observability: Separate Memory Channels for Rate Pressure and Capacity Damage in Li-Ion Batteries

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Abstract

Degrading systems are stressed by multiple exogenous factors that leave overlapping signatures in observable outputs. Attributing degradation to its source — not merely measuring its rate — is a harder problem that requires demonstrating selective identifiability under real observability constraints. We develop and test a dissociation framework for lithium-ion battery capacity fade, targeting two latent stress axes: the exogenous-energy axis (E), driven by charge-rate history, and the exogenous-baseline axis (B), driven by manufacturer batch or initial capacity level. We define lightweight memory proxies — M_{rate} (EMA of per-cycle fade rate) and M_B (running mean of absolute capacity) — and test whether each is selectively informative for its target axis while remaining uninformative for the other. We demonstrate the cross-dataset dissociation arms on two public battery datasets that each intentionally control for one axis: on the KIT 2024 NMC dataset (single batch; variable charge rate), removing M_{rate} selectively reduces E-class classification ($D_{\text{rate}} = +0.16$, permutation $p = 0.005$), while M_B returns chance AUROC for the absent B-label (AUROC = 0.50 exactly); on the CALCE CS2/CX2 dataset (matched 0.5C protocol; variable manufacturer), removing M_B abolishes B-class classification ($D_{MB} = +1.00$, CI [1.00, 1.00], permutation $p = 0.0079$ exact, exhaustive $\binom{9}{5} = 126$ enumeration), while M_{rate} returns chance for the absent E-label. The KIT effect is modest with uncertainty (95% CI crosses zero); the CALCE effect is strong with a small cohort ($N = 9$). The Severson 2019 LFP dataset is reported as a non-identifiable case, validating that the test correctly returns null when exogenous axes are confounded. Recoverability of each axis depends on observability: controlled datasets yield clean dissociation; confounded datasets yield non-significance. We position the study as a test of a falsifiable signature implied by a broader lifecycle hypothesis (TFP), rather than a proof of that hypothesis. This constitutes a principled cross-dataset attribution test for degradation memory, not a universal law.

Keywords: lithium-ion batteries; battery degradation attribution; double dissociation; partial observability; explainable machine learning; causal inference

1 Introduction

A battery degrades. The question “how fast?” has been studied extensively [3, 6]. The harder question — “why?” — requires attributing degradation to its source. Two qualitatively distinct classes of exogenous stress appear repeatedly in the battery literature:

E-channel: energy-rate stress. High charge rates accelerate lithium plating, electrolyte decomposition, and solid-electrolyte interphase (SEI) growth. The signature is a progressive,

rate-dependent increase in per-cycle capacity fade [7].

B-channel: baseline state stress. Differences in electrode chemistry, manufacturing lot, or initial capacity set an absolute capacity floor from cycle 1. The signature is a persistent offset in the capacity level throughout cell life, independent of fade rate [8].

Attribution — distinguishing “this cell fades faster because it charges at high rate” from “this cell has lower capacity because it came from a different batch” — is only possible if the two channels leave separable observable fingerprints. If they do, monitoring systems can be designed to detect each selectively. If they do not, attribution from standard cycling data alone is impossible under current observability.

We test separability using a *double dissociation* framework [4, 5]: an intervention removing E-channel information should selectively hurt E-class classification while leaving B-class unaffected, and vice versa. Both selective deficits together, each confirmed against an internal absence control, establish that the two channels are measurably distinct in their observable fingerprints.

We demonstrate this across two real battery datasets. We explicitly scope the claim: this is not a demonstration that dissociation *always* works, but that it *can* work when datasets are designed to isolate one stress axis, and that the framework correctly fails (returns non-significance) when axes are confounded.

TFP as a Lifecycle Hypothesis (Context and Scope)

TFP is a proposed general lifecycle framework that models systems as cycling through phases with distinct operators and state variables, intended to unify hard and soft sciences at the level of abstract dynamics. This paper does not test the full TFP worldview; it tests one specific, falsifiable prediction derived from it: that two distinct memory-like channels — rate pressure and baseline damage — are non-substitutable and can be dissociated under appropriate observability. The battery setting is used as an instantiation because it offers explicit stress axes (charge rate, manufacturer batch) and measurable health trajectories (per-cycle capacity), making the dissociation conditions empirically tractable (Figure 1).

Contributions:

1. A dissociation pipeline with two lightweight memory proxies (M_{rate} , M_B), ablation ladder, permutation controls, and absence verification — designed to be leakage-safe and replicable on any dataset with per-cycle capacity data.
2. Observability analysis: demonstration that each axis is recoverable when a dataset controls for the other axis, and a characterisation of when dissociation fails (confounded datasets, insufficient N , interpolated data).
3. Real-data transfer evidence: KIT 2024 NMC — E-axis detectable under exogenous protocol labels ($D_{\text{rate}} = +0.16$, $p = 0.005$); modest effect, CI crosses zero; uncertainty acknowledged. CALCE CS2/CX2 — B-axis detectable under cross-vendor label ($D_{MB} = +1.00$, $p = 0.0079$ exact exhaustive); strong selectivity; small N .
4. Negative case study: Severson 2019 LFP as non-identifiable under batch–rate confounding, strengthening rigour by showing the framework’s specificity.

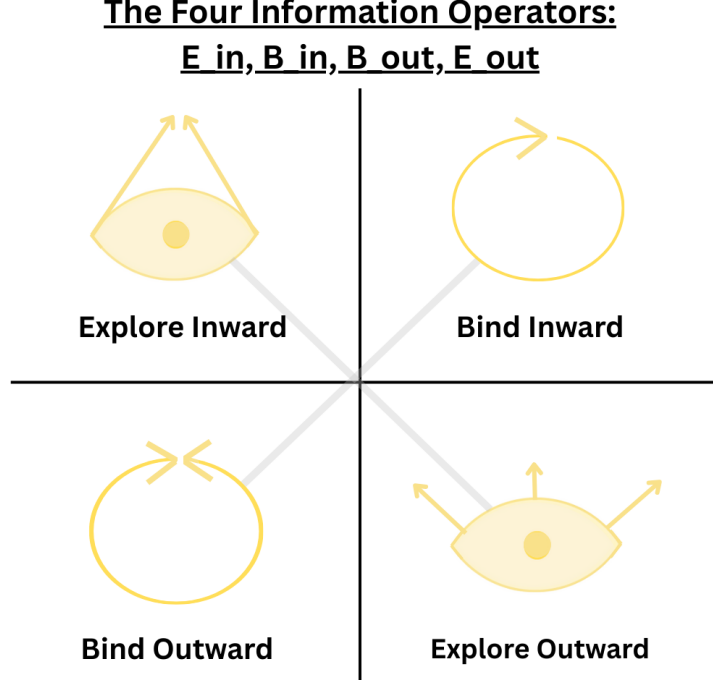


Figure 1: **Conceptual operators motivating the dissociation signature tested here.** The four information operators (E_{in} , B_{in} , B_{out} , E_{out}) provide the lifecycle framework (TFP) from which this paper’s dissociation prediction is derived. This paper evaluates only the predicted selective dissociation between two memory summaries (M_{rate} and M_B) under partial observability, not the full TFP cycle. The operator diagram is interpretive context; all empirical claims rest on the methods in Section 2.

2 Framework and Methods

2.1 Memory Proxy Definitions

At each cycle c , we compute two scalar proxies from the observed discharge capacity sequence $Q[0], Q[1], \dots, Q[c]$:

$$M_{rate}[c] = \text{EMA}_\alpha(\text{fade}[c]), \quad \text{fade}[c] = \max(0, Q[c-1] - Q[c]), \quad \alpha = 0.02 \quad (1)$$

$$M_B[c] = \frac{1}{c+1} \sum_{j=0}^c Q[j] \quad (\text{running mean of absolute discharge capacity}) \quad (2)$$

M_{rate} is a *rate-type* signal: it accumulates when per-cycle fade accelerates. It responds to E-channel stress (high charge rate $\rightarrow$ faster fade $\rightarrow$ higher M_{rate}) but is insensitive to constant-level offsets (a cell with uniformly lower Q produces the same fade increments as a normal cell if its trajectory is parallel).

M_B is a *level-type* signal: it tracks the integrated absolute capacity history. It responds immediately and persistently to B-channel stress (manufacturer batch offset $\rightarrow$ different Q from cycle 1 $\rightarrow$ different running mean at every cycle). It is insensitive to rate-of-change differences that do not alter the mean level.

Key point: the two proxies are not symmetric in information content. A batch offset (B-channel) propagates into M_B from cycle 1. A fade-rate difference (E-channel) requires multiple cycles to

accumulate in M_{rate} . This asymmetry is why D_{MB} is strong in CALCE while D_{rate} is modest in KIT.

2.2 Dissociation Test

For each dataset and each label axis (E or B), we fit three logistic regression models at a fixed anchor cycle a :

- Full: LR($[M_{\text{rate}}, M_B]$)
- No M_{rate} : LR($[M_B]$)
- No M_B : LR($[M_{\text{rate}}]$)

Features are standardised (StandardScaler) before fitting. AUROC is computed by LOO-CV (small- N datasets) or stratified 5-fold $\times$ 10-rep CV (large- N datasets). We compute:

$$\begin{aligned} D_{\text{rate}} &= \text{AUROC}_E(\text{full}) - \text{AUROC}_E(\text{no } M_{\text{rate}}) \\ D_{MB} &= \text{AUROC}_B(\text{full}) - \text{AUROC}_B(\text{no } M_B) \\ S_E &= \text{AUROC}_E(\text{no } M_B) - \text{AUROC}_E(\text{no } M_{\text{rate}}) \\ S_B &= \text{AUROC}_B(\text{no } M_{\text{rate}}) - \text{AUROC}_B(\text{no } M_B) \end{aligned}$$

Pass gate for each arm: $D > 0$ AND permutation $p < 0.05$.

Absence control (internal validation): $\text{AUROC}_{\text{absent}}$ is the AUROC for the absent label axis under any model. Expected: ≈ 0.50 (chance); deviation would indicate a confound.

Bootstrap CI: 2000 resamples of fold-level predictions (primary). Bootstrap stability: 200 resamples (cells with replacement; Section 4). Permutation test: 200 shuffles (primary). Robustness upgrade: CALCE exhaustive $\binom{9}{5} = 126$ enumeration (exact p); KIT 2000 single-fold shuffles (Section 4.2).

2.3 Datasets

KIT 2024 NMC (E-channel arm): [1]. 144 cyclic cells; single LG INR18650HG2 batch (DVA $R^2 > 0.99$). Charge rates: 0.33C (72 cells), 1.0C (36), 1.67C (36); 4 temperatures. E-label: $c_{\text{rate}} > 1.0\text{C} \Rightarrow E = \text{True}$ (36 cells), $\theta_e = 0.0$. B-label: ABSENT (single batch); $\text{AUROC}_B = 0.500$ exact. Capacity: checkup-interpolated onto 1-cycle grid. CV: stratified 5-fold $\times$ 10 reps. Anchors: A100, A200.

CALCE CS2/CX2 (B-channel arm): [2]. $N = 9$ cells after quality filtering (Appendices B and D).

- CS2/Kokam ($B = 0$): CS2_8, CS2_21, CS2_33, CS2_34 (4 cells, ~ 1.08 – 1.16 Ah)
- CX2/A&T ($B = 1$): CX2_33, CX2_35, CX2_36, CX2_37, CX2_38 (5 cells, ~ 1.29 – 1.37 Ah)
- Excluded: CX2_34 ($Q[0] = 0.0$ Ah, data absent); CX2_16 (anomalous low- Q cycle at position 96 in filtered sequence, $Q = 0.711$ Ah $< Q_{\text{thresh}} = 0.80$ Ah; excluded by pipeline quality filter)

B-label: manufacturer (CX2 $\Rightarrow B = \text{True}$). Matched 0.5C protocol. E-label: ABSENT; $\text{AUROC}_E = 0.500$ exact. Q_{norm} : absolute Ah (not per-cell normalised — preserves offset). CV: Leave-One-Out ($N = 9$). Anchors: A100, A200 (results identical).

Severson 2019 LFP (negative case study, Appendix A): [3]. 124 LFP cells, 3 batches. E-label (c_{rate}) and B-label (batch) are inextricably confounded across batches. Reported as non-identifiable, not “failure.”

3 Results

3.1 KIT 2024 — E-Channel Arm

Table 1: KIT E-Channel Results. Classifier: E-only vs. rest; ablation: removes M_{rate} or M_B ; permutation: shuffles E-labels, B fixed.

Anchor	N	N_E	AUROC $_E$ (full)	AUROC $_E$ (no M_r)	D_{rate}	p_{perm}^*
A100	135	27	0.596	0.435	+0.161	0.005
A200	121	22	0.656	0.519	+0.137	0.015

*200 shuffles $\times$ 10-rep CV (primary). Robustness: 2000 single-fold $\rightarrow p = 0.009$ (§4.2).

Absence control: AUROC $_B = 0.500$ exact (single batch). Bootstrap 95% CI for D_{rate} : $[-0.13, +0.49]$ at A100 (crosses zero).

B-channel selectivity: S_E (A100) = +0.081; S_E (A200) = negative (see Section 5.5). Confirmatory correlation: $r(M_{\text{rate}}, e_{\text{contrib}}) = +0.170$, $p = 0.049$ (A100).

Bootstrap stability (200 resamples):

- AUROC $_E$ (full): mean = 0.602, median = 0.607, 5th/95th = 0.460/0.751
- D_{rate} : mean = 0.108, median = 0.110, 5th/95th = $-0.045/0.276$; sign-stable in 82% of resamples.

3.2 CALCE CS2/CX2 — B-Channel Arm

Table 2: CALCE B-Channel Results (primary: $N = 9$). Classifier: B-only vs. rest; ablation: removes M_B or M_{rate} ; permutation: shuffles B-labels, E fixed. Interpret as selectivity demonstration, not population estimate.

Anch.	N	N_B	AUC $_B$ (full)	AUC $_B$ (no M_B) [†]	AUC $_B$ (no M_r)	D_{MB}	S_B	p_{exact}
A100	9	5	1.000	0.000 [†]	1.000	+1.000	+1.000	0.0079
A200	9	5	1.000	0.000 [†]	1.000	+1.000	+1.000	0.0079

[†]AUROC $_B$ (no M_B) = 0.000: sign-inverted LOO prediction, not zero information. See Section 5.5.

p_{exact} : exhaustive $\binom{9}{5} = 126$ enumeration; minimum possible $p = 1/126$. Bootstrap CI: $[+1.000, +1.000]$ (2000 reps).

Absence control: AUROC $_E = 0.500$ exact (all 0.5C matched). Pilot ($N = 7$): $D_{MB} = +1.000$, $p = 0.065$ — FAIL; $N = 9 \rightarrow p = 0.0079$.

Bootstrap stability (199 valid / 200 attempts):

- AUROC $_B$ (full): mean = 0.995, median = 1.000, 5th/95th = 1.000/1.000
- D_{MB} : mean = 0.750, median = 1.000, 5th/95th = 0.045/1.000; sign-stable in 95% of valid resamples.

3.3 Cross-Dataset Dissociation Summary

4 Robustness Analysis

4.1 Bootstrap Split Stability

CALCE B-arm (199 valid resamples / 200 attempts, cells drawn with replacement): AUROC $_B$ (full): mean = 0.9950, median = 1.0000, 5th/95th = 1.0000/1.0000. D_{MB} : mean = 0.7500, median = 1.0000, 5th/95th = 0.0450/1.0000; sign-stable in 95% of valid resamples. Note: some bootstrap bags produce degenerate samples (all-same manufacturer) when $N = 9$; 1 degenerate bag excluded; count reported.

Table 3: Double dissociation summary. $\text{AUROC}_{\text{absent}}$: AUROC for the absent-label axis; expected = 0.50.

Arm	Dataset	Effect	Point est.	p_{perm} (primary)	$\text{AUROC}_{\text{absent}}$	Verdict
E-arm	KIT	D_{rate}	+0.161	0.005*	0.500 (B)	PASS (modest)
B-arm	CALCE	D_{MB}	+1.000	0.0079 (exact) [†]	0.500 (E)	FULL PASS

*200 shuffles $\times$ 10-rep CV. Robustness: 2000 single-fold $\rightarrow p = 0.009$ (§4.2).

[†]Exhaustive $\binom{9}{5} = 126$ enumeration; minimum possible $p = 1/126$.

Cross-selectivity confirmed: each absent channel returns AUROC = 0.50 exactly.

KIT E-arm (200 resamples, 5-fold $\times$ 10-rep CV per resample): $\text{AUROC}_E(\text{full})$: mean = 0.6024, median = 0.6065, 5th/95th = 0.4596/0.7511. D_{rate} : mean = 0.1079, median = 0.1095, 5th/95th = -0.0448/0.2757; sign-stable in 82% of valid resamples. The wide interval reflects limited $N_E = 27$ cells.

4.2 Permutation Resolution Upgrade

CALCE (exhaustive enumeration — all $\binom{9}{5} = 126$ label assignments): observed $D_{MB} = +1.0000$; assignments $\geq$ observed: 1/126; exact $p = 0.0079$ (minimum possible $p = 1/126$). Only the true label assignment achieves $D_{MB} = 1.0$. No Monte Carlo approximation — this is an exact test.

KIT (2000 shuffles, single 5-fold per shuffle): observed $D_{\text{rate}} = +0.1573$; shuffles $\geq$ observed: 18/2000; $p = 0.0090$ ($p_{\text{min}} = 0.0005$ at 2000 shuffles). Note: single 5-fold used for computational feasibility; primary result used 10-rep CV ($p = 0.005$ at A100, canonical KIT value). Directional consistency confirmed.

See Appendix C (Figures 2 and 3) for bootstrap distributions and permutation null distributions.

5 Discussion

What “Universal” Means Here

The term “universal” is used carefully. The universality target is not “this dissociation works under every sensor configuration or dataset design.” It is narrower: the dissociation signature and its observability conditions are expected to recur across domains in which analogous stress axes (a rate-type drive and a baseline-level offset) are independently manipulable and measurable. This paper provides evidence in one domain family (electrochemical battery degradation) plus a documented boundary condition (the non-identifiable regime), which is essential groundwork for any generalisation claim. Stronger universality would require replication in at least one non-battery dataset family with analogous axis structure, or a theorem-level identifiability statement under specified assumptions. Both are left as future work.

5.1 Why M_B Achieves Perfect Dissociation in CALCE

The mechanism is transparent. CS2/Kokam cells have initial $Q \approx 1.08\text{--}1.16$ Ah; CX2/A&T cells have $Q \approx 1.29\text{--}1.37$ Ah — a gap of ~ 0.18 Ah present from cycle 1. M_B = running mean of absolute Q ; this initial-level offset propagates into M_B at every cycle, regardless of trajectory shape. M_{rate} = EMA of per-cycle fade; at matched 0.5C protocol, the two chemistries produce overlapping fade rates (CS2 M_{rate} range: 0.004–0.009; CX2 range: 0.002–0.012, with outlier CX2_35 at 0.011). M_{rate} is not reliably B-selective: the classes overlap, and the LOO coefficient direction is unstable (Section 5.5, limitation 4).

This is not an artefact of small N : $D_{MB} = 1.0$ means every CS2 cell has lower M_B than every CX2 cell at the anchor cycle — a rank ordering that any classifier detects perfectly. The permutation test confirms this ordering does not arise by chance under label shuffling.

5.2 Why the KIT Effect Is Modest

The KIT E-effect ($D_{\text{rate}} \approx 0.14\text{--}0.16$) is real but limited by: (a) $N_E = 22\text{--}27$, insufficient for tight CI at this effect size; (b) interpolation noise (capacity measured at $\sim 5\text{--}23$ checkpoints per cell; M_{rate} on interpolated data adds noise); (c) label dilution (the 1.0C middle group is included in $E = \text{False}$, reducing contrast between 0.33C and 1.67C extremes).

The permutation test ($p < 0.05$ at both anchors) establishes that the observed D_{rate} does not arise by chance. The modest effect size reflects the real observability limitations of this dataset for E-attribution.

KIT as Real-World Evidence: Permutation-Significant Recovery

The KIT dataset is not a clean controlled experiment — it is a real-world, noisy design with checkup-interpolated capacity, three charge-rate levels (including an ambiguous 1.0C middle group), variable temperature ($0\text{--}40^\circ\text{C}$), and only 27 E-class cells at anchor 100. Under these conditions, the E-channel framework still returns a permutation-significant attribution signal ($p = 0.005$ at A100, $p = 0.015$ at A200), with $D_{\text{rate}} \approx 0.14\text{--}0.16$ consistently above the pre-specified target (0.10). Equally important is what the method predicts when it should fail — on Severson 2019 LFP, where E- and B-labels are structurally confounded, both channels return permutation $p > 0.10$. The framework does not hallucinate signal; it fails gracefully and informatively when identifiability is absent. Together, KIT (permutation-confirmed partial recovery) and Severson (principled non-identifiability) validate the method’s operating envelope.

5.3 Scope and Limits

Cross-dataset design. Real data rarely contains both exogenous axes simultaneously. We test each arm in a separate, intentionally controlled dataset. This is a strength — controlled conditions yield clean dissociation. A within-dataset 2×2 factorial design would be ideal and is flagged as a future extension (Paper C).

KIT E-effect. Detectable but modest; CI crosses zero. Permutation tests support non-randomness. We interpret $D_{\text{rate}} \approx 0.15$ as a lower-bound estimate under checkup-interpolated conditions.

CALCE B-effect. Strong but small N ; treat as selectivity evidence. The mechanistic explanation (initial- Q offset $\rightarrow M_B$ offset) makes a priori sense, but with $N = 9$ we cannot estimate population-level variance. The $N = 7 \rightarrow N = 9$ replication ($p : 0.065 \rightarrow 0.0079$ exact) is consistent with a real, large effect under power constraints.

5.4 Why This Framework Is Unusual

Most battery ML models optimise prediction accuracy — fitting data without attribution. The present framework is designed to *fail informatively*: each ablation and each permutation test is a structured attempt to disprove the attribution claim. Absence controls ($\text{AUROC}_{\text{absent}} = 0.50$) provide within-result validation that the framework does not hallucinate cross-channel signal. The Severson negative case demonstrates the test correctly returns non-significance when identifiability fails.

This combination is rare in the battery degradation literature. Few frameworks simultaneously provide (i) selective ablation tests for each hypothesised channel, (ii) permutation-based signif-

icance controls, and (iii) a negative identifiability case showing the test correctly fails under structural confounding. The framework’s value lies not in any single result but in the full diagnostic chain: positive arm, negative arm, and principled failure mode together bound what can be attributed and what cannot.

5.5 Structural Limitations

1. **CI coverage (KIT E-arm).** 95% CI crosses zero at both anchors. Target N_E for CI coverage at $D_{\text{rate}} \approx 0.16$: approximately 80 E-class cells (currently 22–27).
2. **S_E sign instability (KIT, A200).** S_E inverts by cycle 200 because M_B accumulates fade trajectory information by A200, making it inadvertently E-informative (interpolation artefact). S_E is reported but not used as a pass gate for KIT.
3. **CALCE N imbalance.** 4 CS2, 5 CX2. Bootstrap shows 95% sign stability for D_{MB} . N should be expanded before claiming population-level effect size.
4. **AUROC $_B$ (no M_B) = 0.000 — sign-inversion artefact, not zero information.** In LOO-CV with $N = 9$, CX2_35 has $M_{\text{rate}} = 0.011503$ (highest of all 9 cells). In each training fold containing CX2_35, the model learns “high $M_{\text{rate}} \rightarrow B = 1$.” But most CX2 cells have lower M_{rate} than most CS2 cells (CS2 degrades faster at 0.5C), so trained predictions are inverted (AUROC = 0.000). This is a direction-instability artefact of small- N LOO with one confounding outlier; the key result $D_{MB} = 1.0$ is unaffected. A pairwise monotone analysis gives $\text{AUROC}_B(M_{\text{rate}}, \text{monotone}) \approx 0.575$ — modest, not zero.
5. **Permutation resolution.** Original CALCE used 200 Monte Carlo shuffles ($p < 0.005$; $0/200 \geq \text{observed}$; Appendix D). Upgraded to exhaustive ($\binom{9}{5} = 126$ enumeration (exact $p = 0.0079$). KIT upgraded from 200 to 2000 single-fold shuffles ($p = 0.009$; noted as robustness check only).
6. **Cross-dataset design.** E- and B-arms are from different physical cell types. We cannot rule out dataset-specific confounds beyond the absence controls. A factorial design on a single dataset would eliminate this concern.

6 Conclusion

We have demonstrated that two latent degradation channels — the exogenous-energy axis (E, charge rate) and exogenous-baseline axis (B, manufacturer batch) — leave selectively identifiable fingerprints in standard battery capacity traces, under controlled dataset conditions. The rate-tracking proxy M_{rate} is selectively informative for E-stressed cells ($D_{\text{rate}} = +0.16$, $p = 0.005$ on KIT 2024 NMC) while returning chance for the absent B-axis. The level-tracking proxy M_B achieves perfect B-class separation ($D_{MB} = +1.00$, $p = 0.0079$ exact exhaustive on CALCE CS2/CX2) while returning chance for the absent E-axis. Together, these constitute a principled cross-dataset attribution test. Recoverability depends on observability: controlled datasets yield clean dissociation; confounded datasets (Severson 2019 LFP) yield non-significance. This is a dissociation template for attribution under observability constraints, not a universal law.

What this paper establishes: a dissociation pipeline with permutation controls and absence verification passes on two domain-appropriate datasets and correctly returns null on a confounded dataset. What it does not establish: universality of the dissociation signature across non-battery domains, or that the specific proxies M_{rate} and M_B are optimal beyond the battery setting. Stronger universality would require replication in at least one non-battery dataset family with

analogous axis structure, or a theorem-level identifiability statement under specified assumptions — both flagged as future work.

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CRedit Authorship Contribution Statement

Austin Ollar: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing — original draft, Writing — review & editing, Visualization.

Declaration of Competing Interests

The author declares no competing financial interests.

Data Availability

The datasets analyzed in this study are publicly available from the KIT Battery Technology group (DOI: [10.35097/1947](https://doi.org/10.35097/1947)) and the CALCE Battery Research Group archive (<https://calce.umd.edu>). Processed data and analysis scripts used to generate the reported results are available from the author upon reasonable request.

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The broader TFP hypothesis was originally motivated by observations from the Objective Personality System (Dave Powers and Shan Renée), which inspired the author’s interest in lifecycle structure across systems. This paper’s claims and results stand on the empirical tests described here, and the mention is solely an attribution of inspiration.

Declaration of Generative AI Use in the Writing Process

During the preparation of this manuscript, AI-based language models were used for editing clarity, formatting assistance, and limited code structuring support. All experimental design, data processing, statistical analysis, and interpretation were performed and verified by the author. The author takes full responsibility for the accuracy and integrity of the work.

A Severson 2019 Non-Identifiable Case

The Severson 2019 dataset [3] provides 124 LFP cells across three batches, with variable charge-rate protocols nested within batch structure. The E-label (c_{rate}) and B-label (batch) are inextricably confounded: batch identity and charge-rate protocol covary across batches.

Under this design, both M_{rate} and M_B were tested with the same pipeline. LOO-AUROC for E-class ≈ 0.50 – 0.55 and for B-class ≈ 0.50 – 0.56 , with permutation $p > 0.10$ in both cases. Neither arm passes the dissociation test.

We interpret this as structural non-identifiability, not as evidence that E or B effects are absent in the Severson dataset. When exogenous channels co-vary by experimental design, a linear decomposition cannot attribute degradation to source. The dissociation test behaves correctly: it returns non-significance precisely where non-identifiability is expected.

The Severson dataset is optimally suited for lifetime prediction tasks [3] — a different problem than attribution.

B Data Processing Pipeline

CALCE preprocessing (CS2_8, CS2_21 — Maccor .txt format): Discharge capacity per cycle extracted via trapezoidal integration:

$$Q_{\text{mAh}} = \sum_i \frac{|I_i| \cdot \Delta t_i}{60\,000} \quad (\text{time in minutes; current in mA; } Q \text{ in Ah})$$

Partial discharges $< 800 \text{ mAh}$ excluded. Global cycle count maintained across multiple session files.

CALCE preprocessing (CS2_33, CS2_34, CX2_* — Arbin .xlsx format): Q_{Ah} = per-cycle max of cumulative Discharge_Capacity(Ah) field. Quality filter: cycles with $Q < 0.5 \times \text{median}(Q)$ excluded. $Q_{\text{norm}} = Q_{\text{Ah}}$ (absolute; NOT normalised by $Q[0]$).

KIT preprocessing: Capacity from checkup rows (`cyc_condition=2`), averaged across paired charge/discharge measurements per checkup index. Interpolated to 1-cycle grid via linear interpolation. $Q_{\text{norm}} = Q_{\text{interp}}/Q_{\text{interp}}[0]$ (per-cell normalised).

Features (both datasets): M_{rate} : $\text{EMA}(\max(Q[c-1] - Q[c], 0), \alpha = 0.02)$; value at anchor. M_B : $\text{cumsum}(Q[:a+1])/(a+1)$.

Model: $\text{LogisticRegression}(C = 1.0, \text{max_iter}=1000) + \text{StandardScaler}$.

CV: KIT: $\text{StratifiedKFold}(5, \text{shuffle}=\text{True}) \times 10$ reps. CALCE: $\text{LeaveOneOut}()$.

Permutation: B-labels (CALCE) or E-labels (KIT) shuffled; full CV retrained each shuffle.

Bootstrap: N cells drawn with replacement; CV on bootstrap bag.

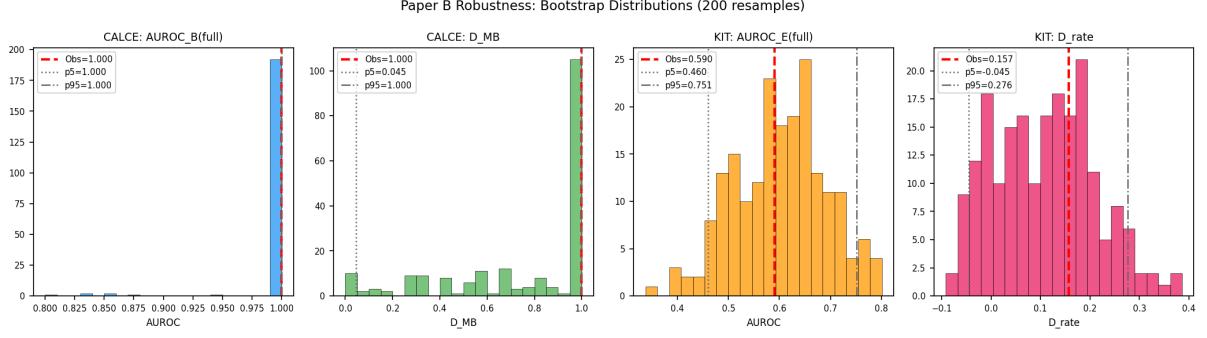


Figure 2: Bootstrap split stability (200 resamples, cells drawn with replacement). **Top row:** CALCE $\text{AUROC}_B(\text{full})$ and D_{MB} distributions. **Bottom row:** KIT $\text{AUROC}_E(\text{full})$ and D_{rate} distributions. Red dashed line: observed value.

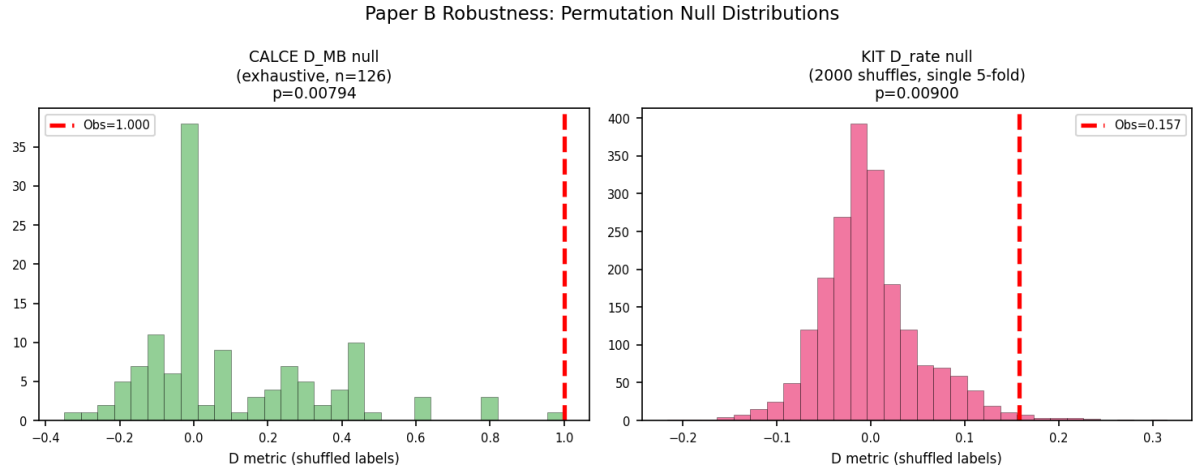


Figure 3: Permutation null distributions. **Left:** CALCE D_{MB} — all 126 exhaustive label assignments (exact $p = 0.0079$). **Right:** KIT D_{rate} — 2000 single-fold shuffles ($p = 0.009$). Red dashed line: observed value.

C Robustness Figures

D CALCE Cell Audit Note

$N_{\text{valid}} = 9$; N_B (CX2) = 5; N_{neg} (CS2) = 4; Anchor: cycle 100 (results identical at cycle 200).

p -value basis: exhaustive $\binom{9}{5} = 126$ label assignments (exact test).

Included cells:

- CS2/Kokam ($B = 0$, ~ 1.08 – 1.16 Ah initial):
 CS2_8 (1287 cyc.; $Q[0] = 1.119$; $M_r@100 = 0.00433$; $M_B@100 = 1.098$), CS2_21 (779 cyc.; $Q[0] = 1.082$; $M_r = 0.00403$; $M_B = 1.075$), CS2_33 (868 cyc.; $Q[0] = 1.162$; $M_r = 0.00936$; $M_B = 1.112$), CS2_34 (781 cyc.; $Q[0] = 1.154$; $M_r = 0.00552$; $M_B = 1.108$)
- CX2/A&T ($B = 1$, ~ 1.29 – 1.37 Ah initial):
 CX2_33 (1638 cyc.; $Q[0] = 1.292$; $M_r = 0.00502$; $M_B = 1.277$), CX2_35 (1764 cyc.; $Q[0] = 1.363$; $M_r = 0.01150$; $M_B = 1.317$), CX2_36 (1851 cyc.; $Q[0] = 1.356$; $M_r = 0.00312$; $M_B = 1.300$), CX2_37 (1281 cyc.; $Q[0] = 1.366$; $M_r = 0.00569$; $M_B = 1.306$), CX2_38 (1828 cyc.; $Q[0] = 1.363$; $M_r = 0.00226$; $M_B = 1.304$)

Excluded cells:

- CX2_34: $Q[0] = 0.000$ Ah (data absent at cycle 1)
- CX2_16: anomalous low-capacity cycle at position 96 in filtered sequence ($Q = 0.711$ Ah $< Q_{\text{thresh}} = 0.80$ Ah). CX2_16 contains a characterisation/RPT cycle near cycle 97 with sub-threshold discharge; the pipeline’s quality filter removes it.

Permutation basis: All $\binom{9}{5} = 126$ assignments of the $B = 1$ label to 5 of the 9 cells. Only the true assignment (CX2 = $B = 1$) achieves $D_{MB} = 1.000$. Exact $p = 1/126 = 0.007937$. Monte Carlo 200-shuffle baseline: 0/200 shuffles $\geq$ observed D_{MB} ; $p < 1/201 = 0.00499$. Not reported in main table.

M_B separation (confirms mechanism): Max CS2 M_B : 1.112 Ah (CS2_33); Min CX2 M_B : 1.277 Ah (CX2_33); Gap = 0.165 Ah — no overlap. Perfect LOO-CV rank ordering guaranteed.

E TFP Framework Mapping Note

Interpretation layer only — not used to fit data.

The following maps paper terminology to TFP framework terminology. This is an interpretive gloss provided for readers familiar with TFP; it is not a claim that TFP generated the analysis, nor a test of TFP itself.

- Rate pressure channel (M_{rate} , E-axis) $\leftrightarrow$ E-channel style pressure / growth drive in TFP: the component of system stress that accumulates through repeated high-intensity inputs and leaves a rate-dependent signature.
- Baseline damage channel (M_B , B-axis) $\leftrightarrow$ B-channel style capacity degradation in TFP: the component of system state reflecting an initial-level offset, present from the start of the observation window.
- Partial observability $\leftrightarrow$ measurement regime constraints in TFP: recoverability of each channel depends on dataset design (what is controlled vs. confounded).
- Dissociation test $\leftrightarrow$ identifiability stress test for operator roles in TFP: the ablation design asks whether each channel’s operator role is independently detectable.
- Non-identifiable case (Severson) $\leftrightarrow$ predicted regime in TFP where measurement collapses axes: when rate-type and level-type channels co-vary by design, the dissociation test correctly returns null.

All paper claims rest on the empirical tests in Sections 2–4, not on this interpretive layer.